

Physical Chemistry of Materials

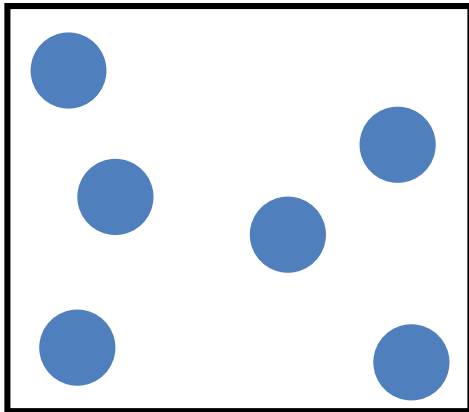
Lecture 18

Liquids and Glasses

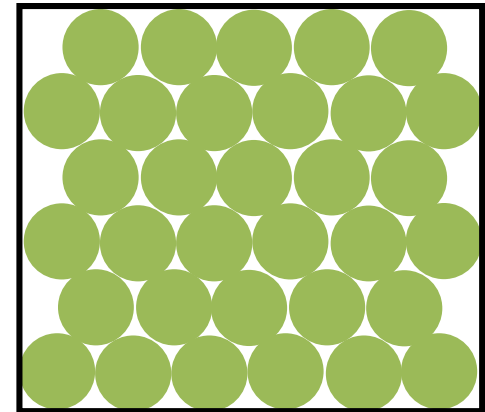
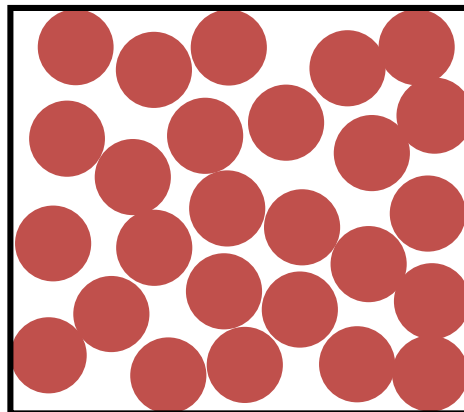


Gas, Liquid, and Solid

condensed phase



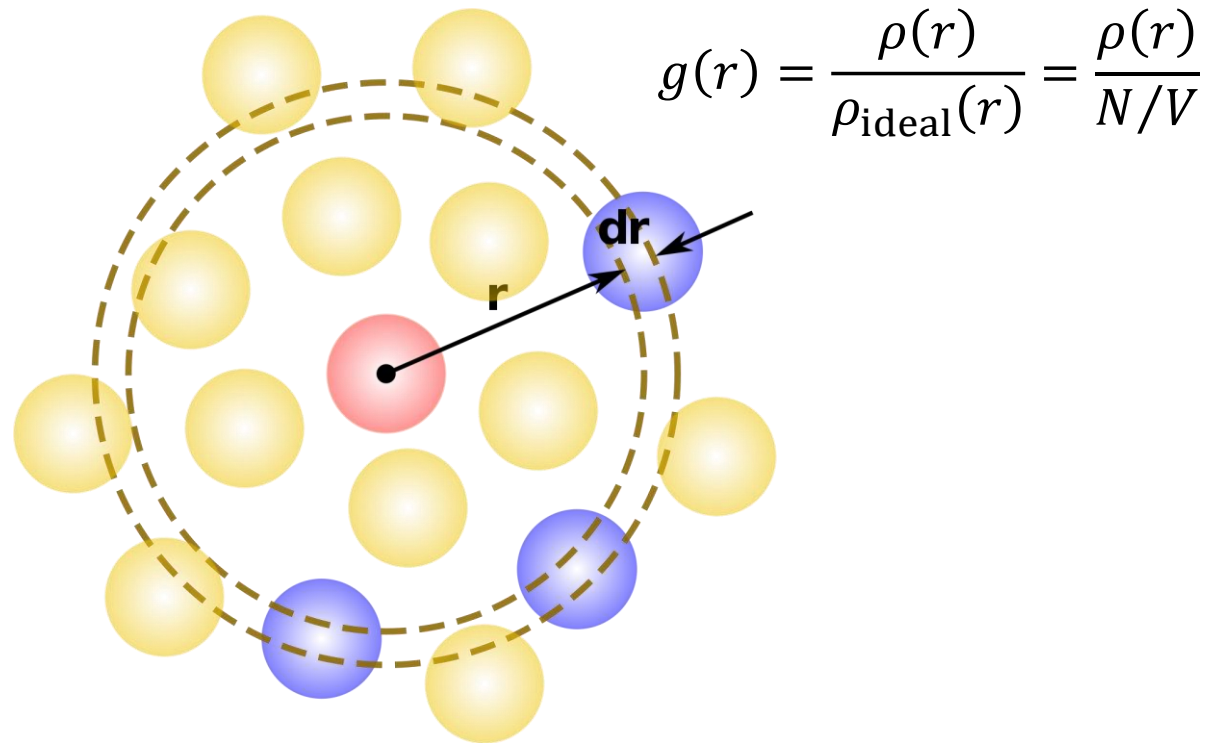
Model: “ideal gas”



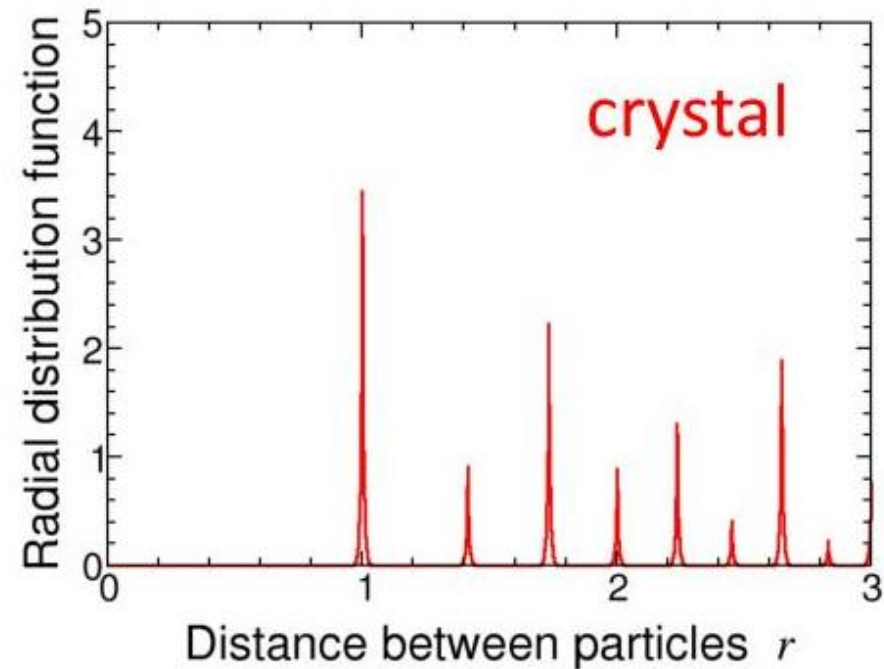
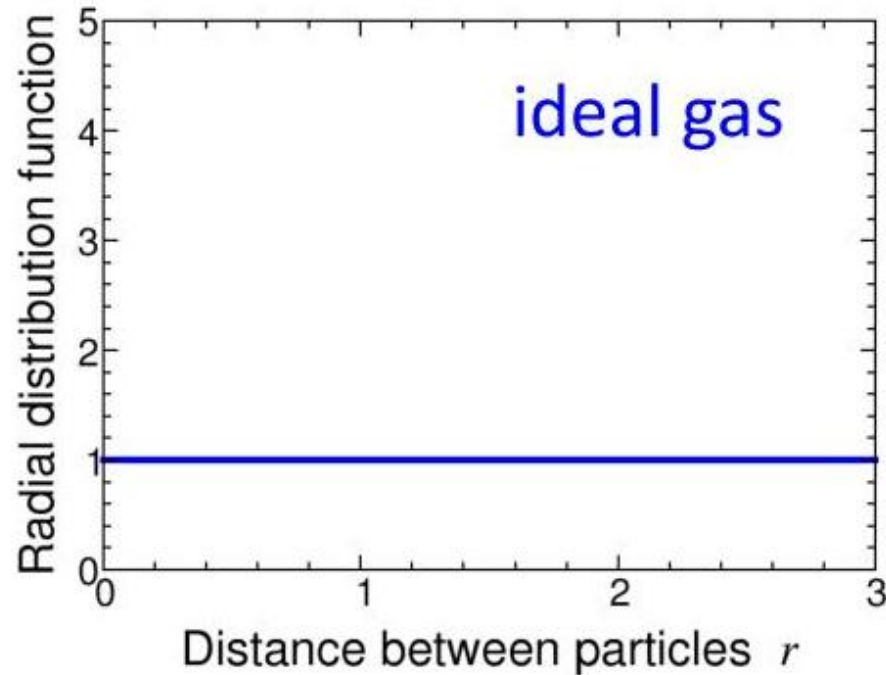
Model: “ideal crystal”

How to quantify the “structure” of a liquid?

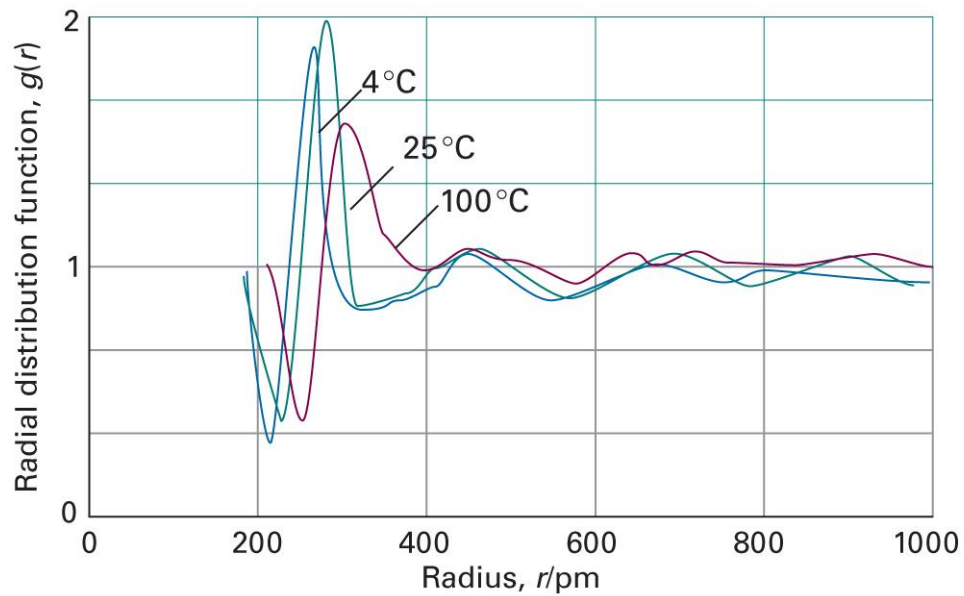
Radial Distribution Function $g(r)$



RDFs of Ideal Gas and Crystal

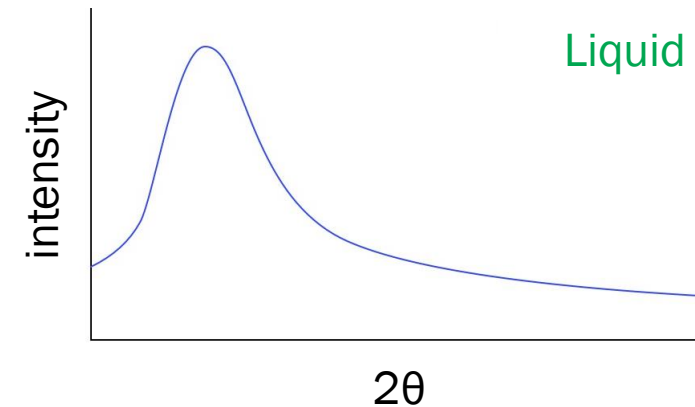
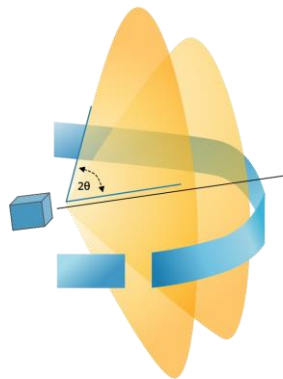
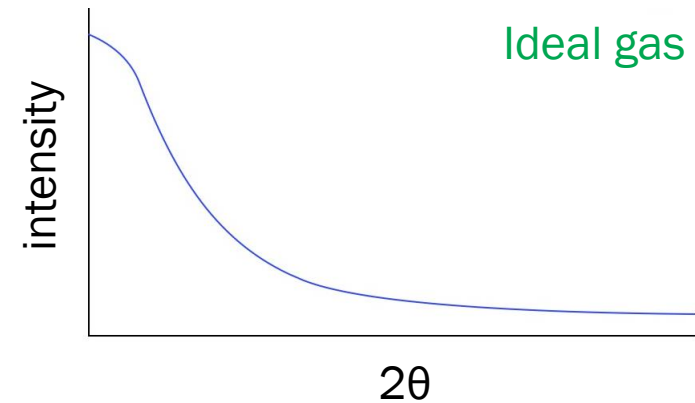
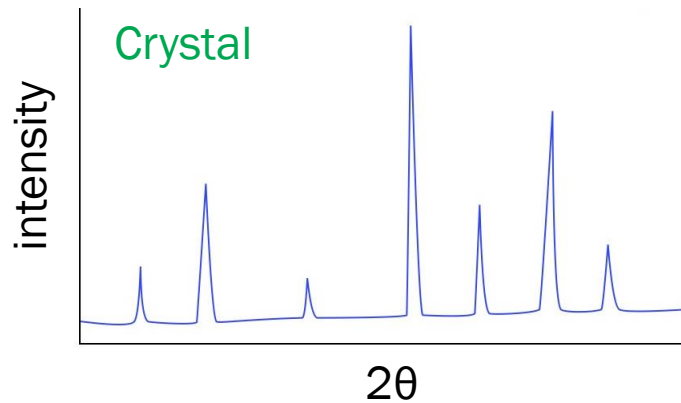


RDF of Liquid Water



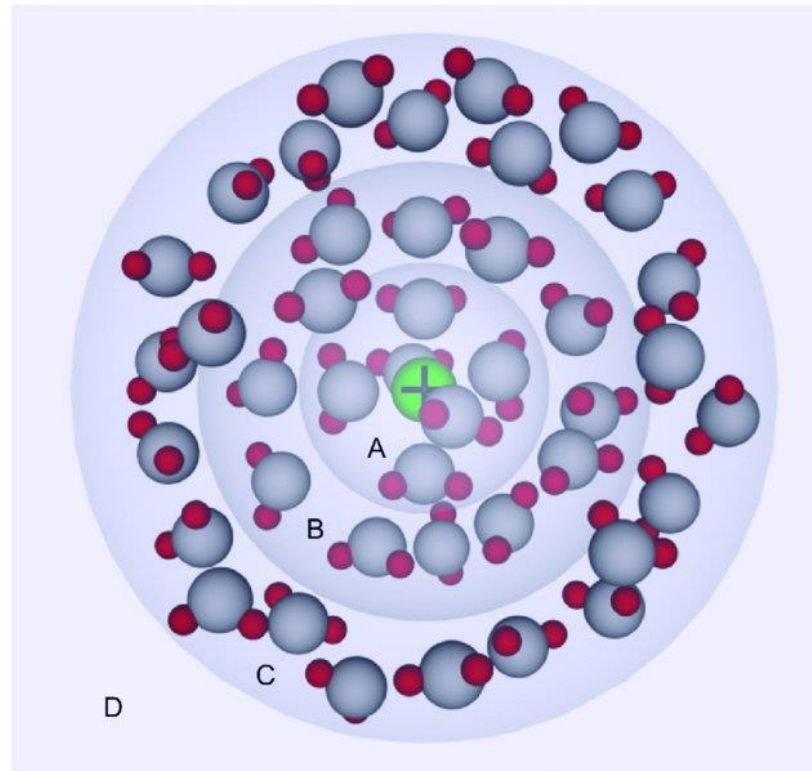
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 14C.1)

Scattering Experiments



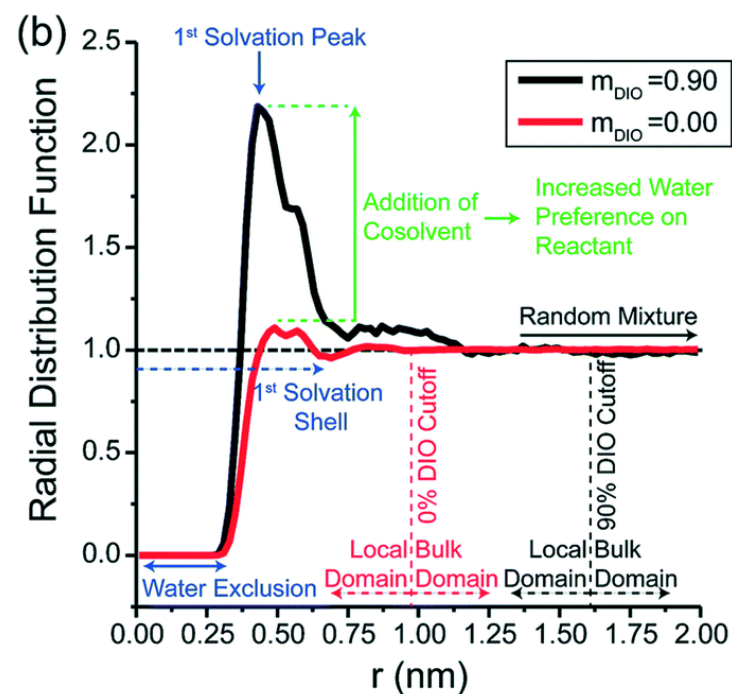
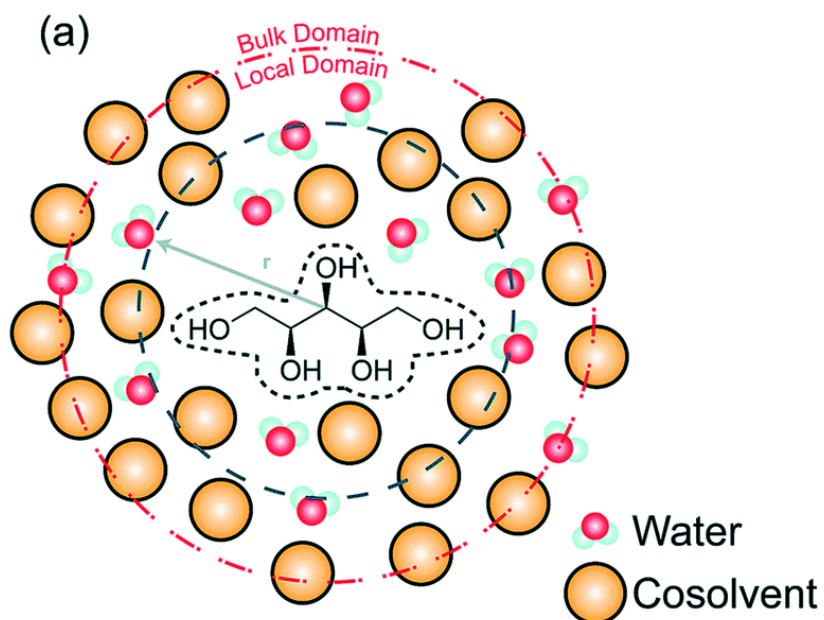
(Image: https://myscope.training/XRD_Diffraction_measurements)

Solvation Shells



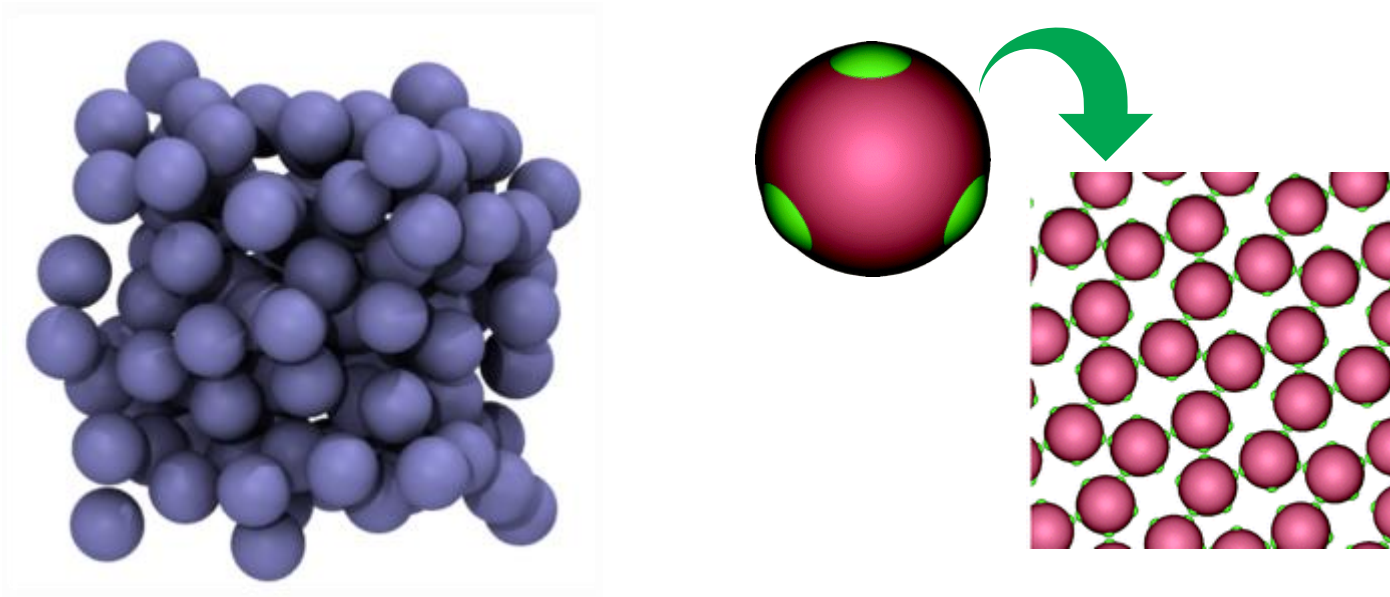
(Image: <https://basicmedicalkey.com/solutions-of-electrolytes/>)

Application of RDF



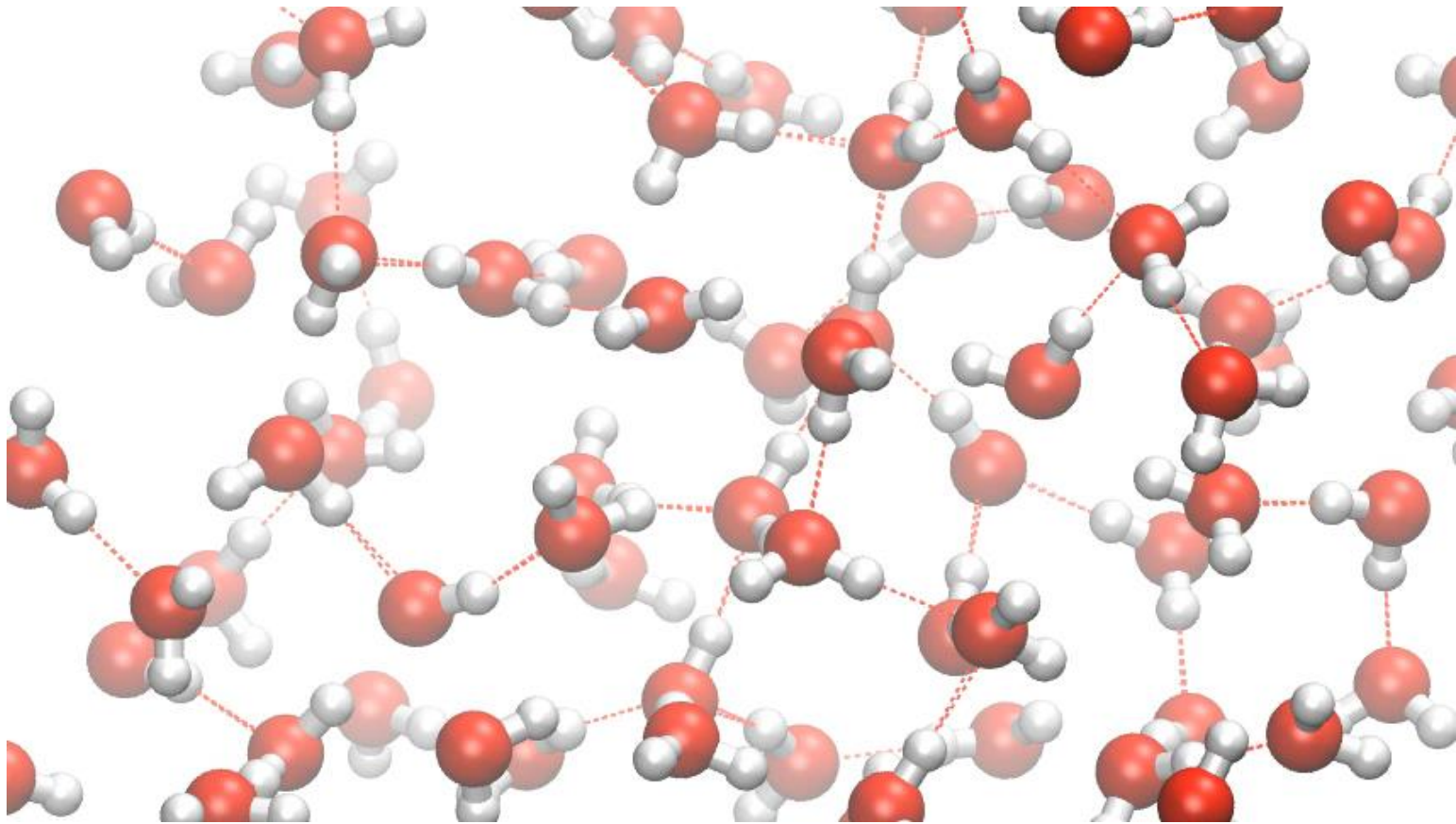
(Image: Varghese and Mushrif, *Reaction Chemistry & Engineering* (2019), Fig. 47)

Simple vs. Network Fluids



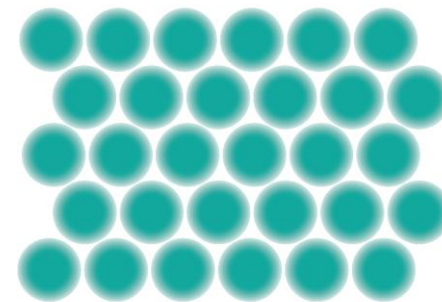
(Images: <https://www.pyretis.org/current/user/getting-started.html>
Bianchi et al., *Phys Chem Chem Phys* (2011))

Structure of Liquid Water

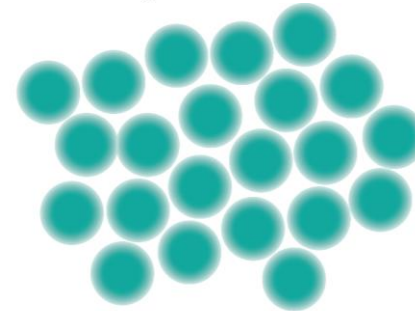


Condensed Phases

- Crystalline solids
 - Amorphous solids
 - Liquids (cf. glasses)
- } ?
- “Soft matters”
 - Polymers
 - Colloids
 - Liquid crystals ...

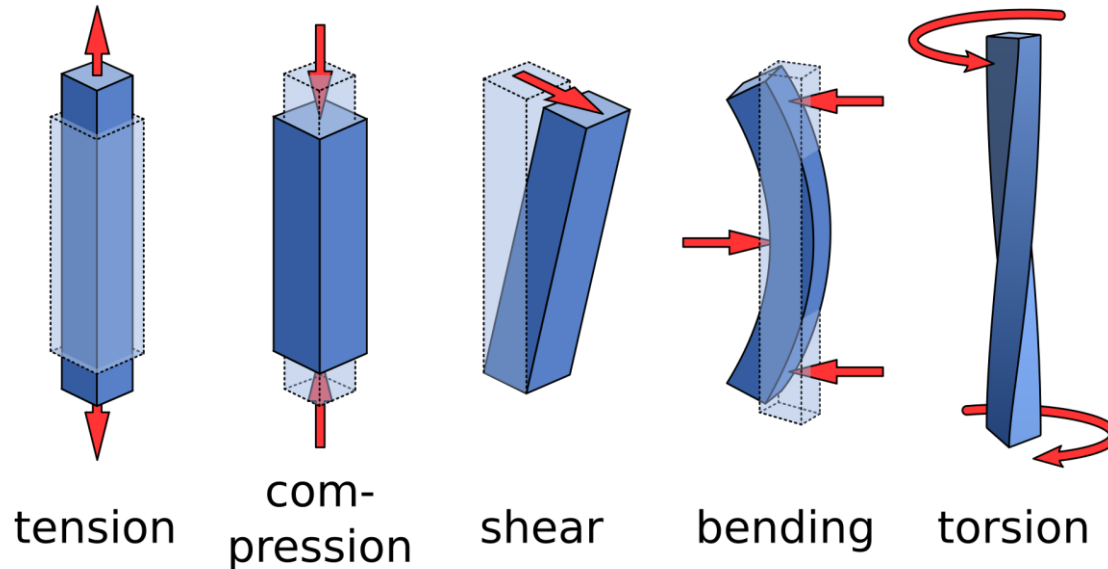


Crystalline



Amorphous

Mechanical Properties of Solids



- **Stress**: (applied force) / (applied area)
- **Strain**: resulting distortion (material's response)

Three Moduli

- Bulk modulus K (all phases)

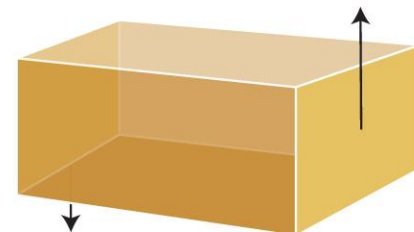
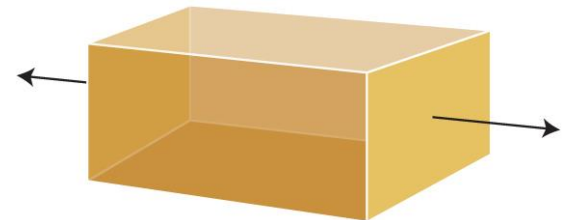
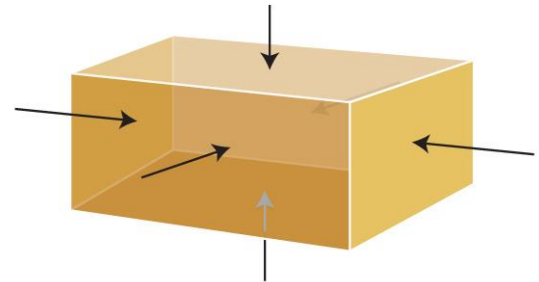
$$K = \frac{\text{pressure}}{\text{relative volume change}}$$

- Young's modulus E (solids)

$$E = \frac{\text{uniaxial stress}}{\text{uniaxial strain}}$$

- Shear modulus G (solids)

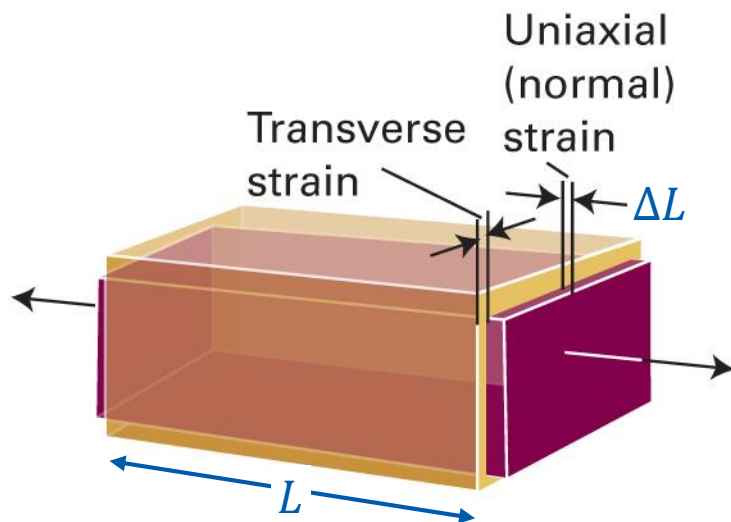
$$G = \frac{\text{shear stress}}{\text{shear strain}}$$



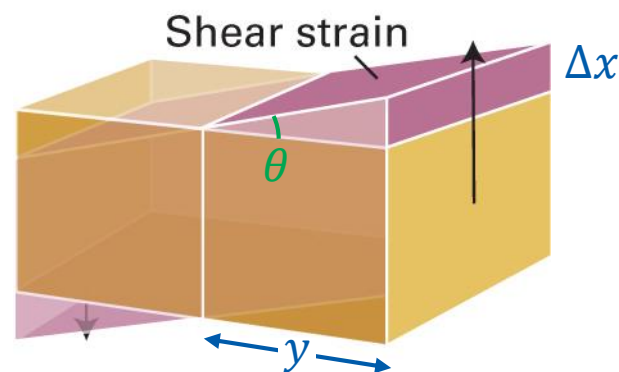
Bulk Modulus of Materials

Material	Phase	Bulk modulus (GPa)
Diamond	Crystalline solid	443
Steel	Crystalline solid	160
Silica glass	Amorphous solid	35-55
Graphite	Crystalline solid	34
Sodium chloride	Crystalline solid	24
Water	Liquid	2.2
Rubber	Amorphous solid	1.5-2
Methanol	Liquid	0.82
Solid helium	Crystalline solid	0.05
Air	Gas	0.0001

Strains: Dimensionless

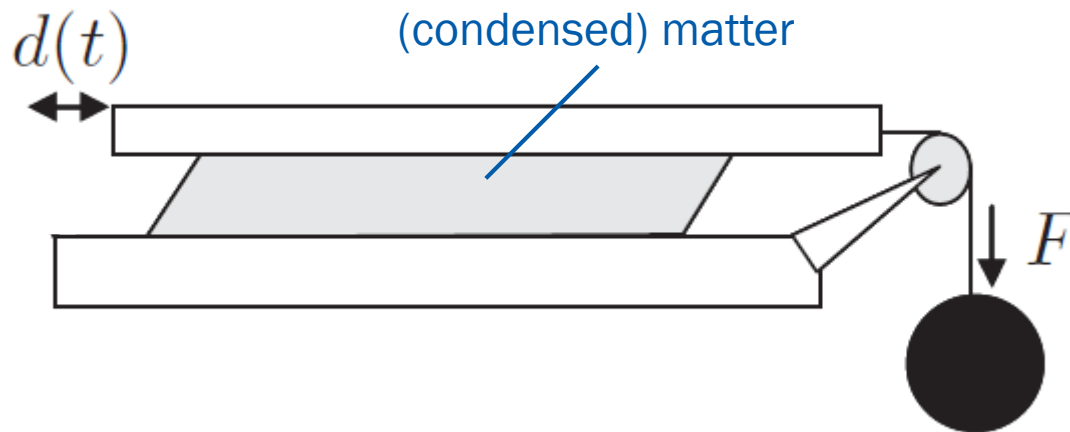


$$\text{uniaxial strain} = \frac{\Delta L}{L}$$



$$\text{shear strain} = \frac{\Delta x}{y} \approx \theta$$

More on Shear Strain



- Ideal fluid keeps flowing at constant rate
- Ideal elastic material moves to the equilibrium position

Elasticity and Viscosity

- Shear stress $\sigma = F/A = (\text{external force})/(\text{applied area})$
- Shear strain $\gamma = d/h = (\text{displacement})/(\text{height})$

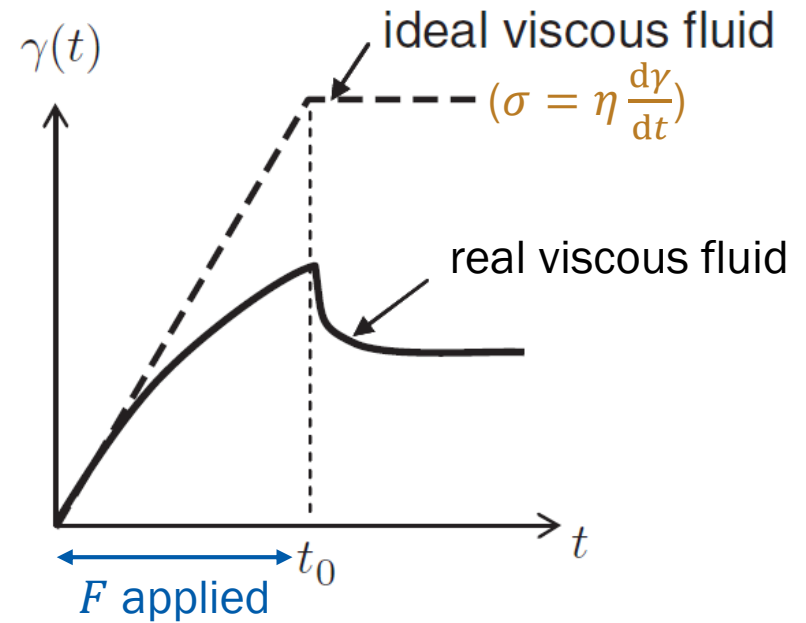
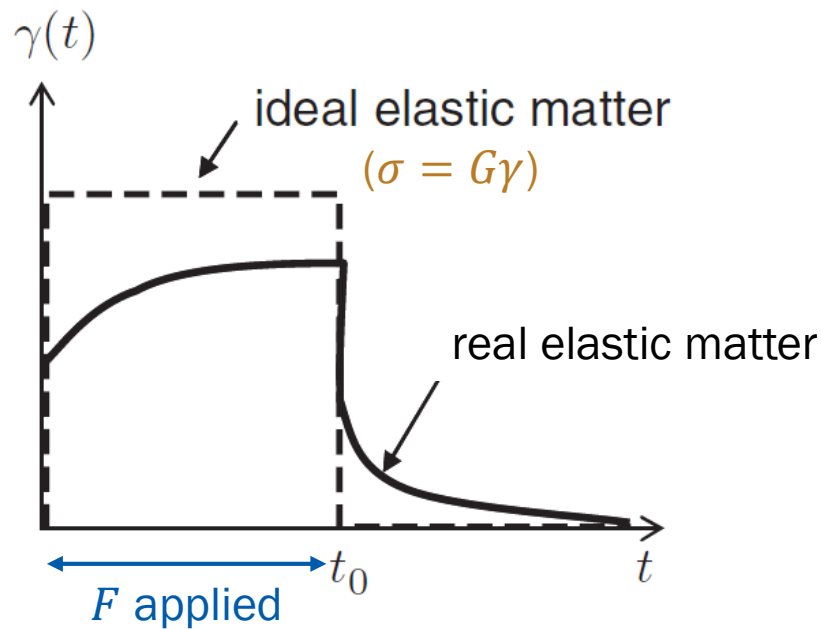
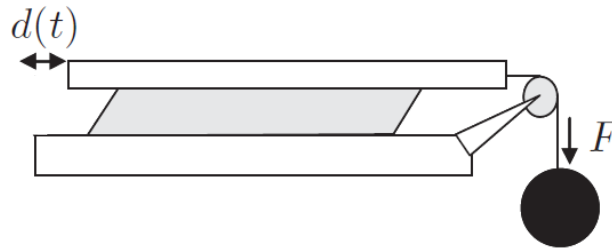
- Ideal elastic material (Hookean solid)

$$\sigma = G\gamma \text{ (} G: \text{shear modulus)}$$

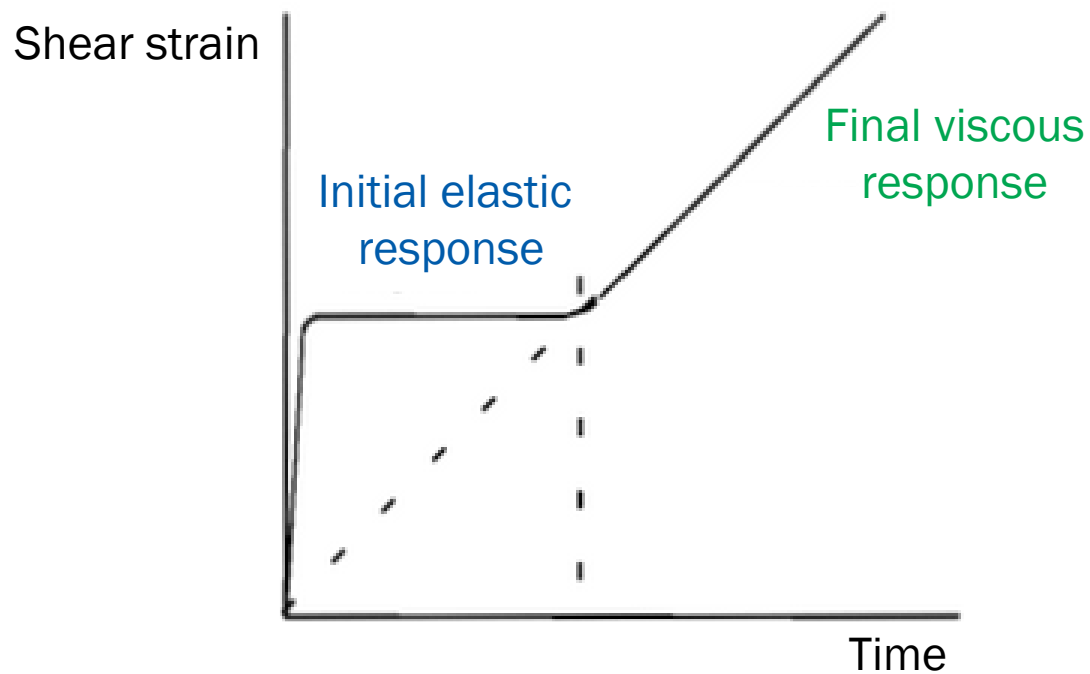
- Ideal fluid (Newtonian fluid)

$$\sigma = \eta \frac{d\gamma}{dt} \text{ (} \eta: \text{viscosity)}$$

Ideal vs. Real



Viscoelastic Materials



(Image: Jones, *Soft Condensed Matter*, Fig. 2.5)

What if $\eta = \eta(\sigma)$?

Flubber (“slime”)



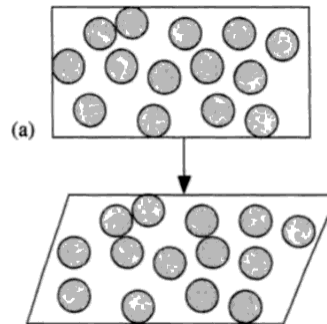
Oobleck



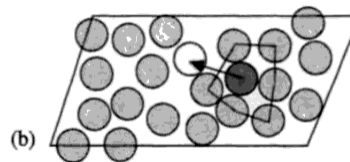
<https://www.youtube.com/watch?v=RIUEZ3AhrVE>

Molecular View on Viscosity

- Instantaneous response of a liquid: no rearrangement



- Relaxation: escape from a “cage” by neighbors



Temperature Dependence

Define an instantaneous modulus G_0 at times much shorter than the relaxation time τ ;

$$\sigma \sim G_0 \gamma = G_0 \tau \frac{\gamma}{\tau} \sim \underbrace{G_0 \tau}_{\eta} \frac{d\gamma}{dt}$$

Relaxation time is related to the “activation energy” E_a ;

$$\tau^{-1} \sim \nu e^{-E_a/RT}$$

Combining the two equations,

$$\eta \sim \frac{G_0}{\nu} e^{E_a/RT} = \eta_0 e^{E_a/RT}$$

Viscosity of Liquids

Mobility $\uparrow \rightarrow$ Viscosity $\downarrow$

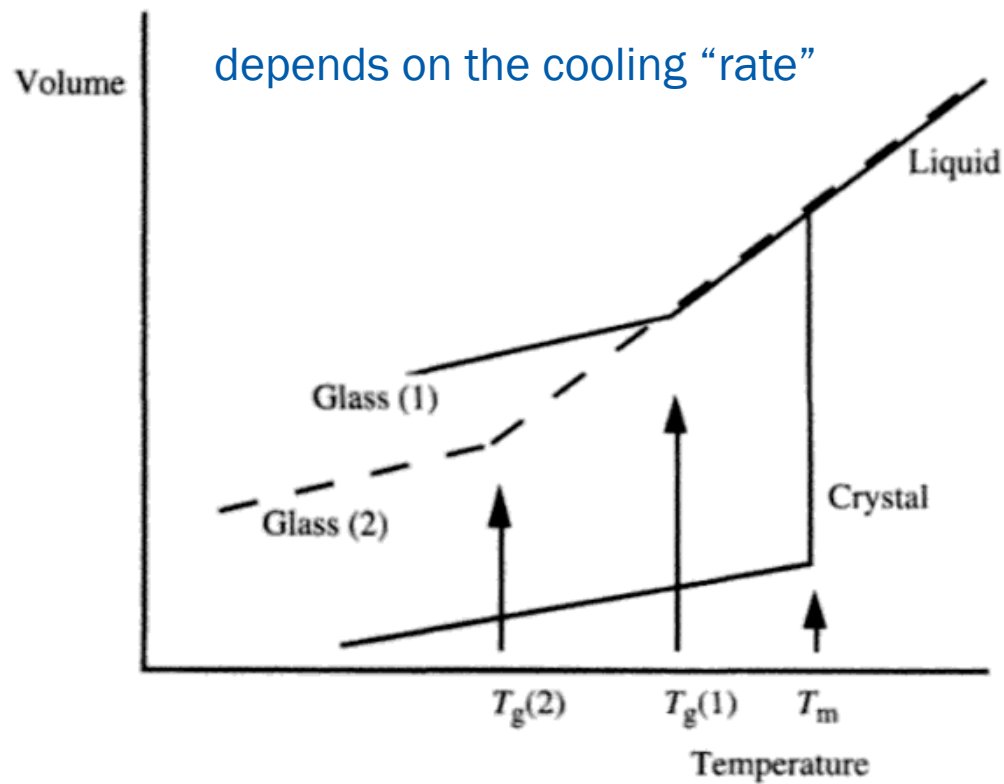
	$\eta/(10^{-3} \text{ kg m}^{-1} \text{ s}^{-1})$
Benzene	0.601
Mercury	1.55
Pentane	0.224
Water	0.891

Glasses

$$\tau \sim \frac{1}{\nu} e^{E_a/RT}$$

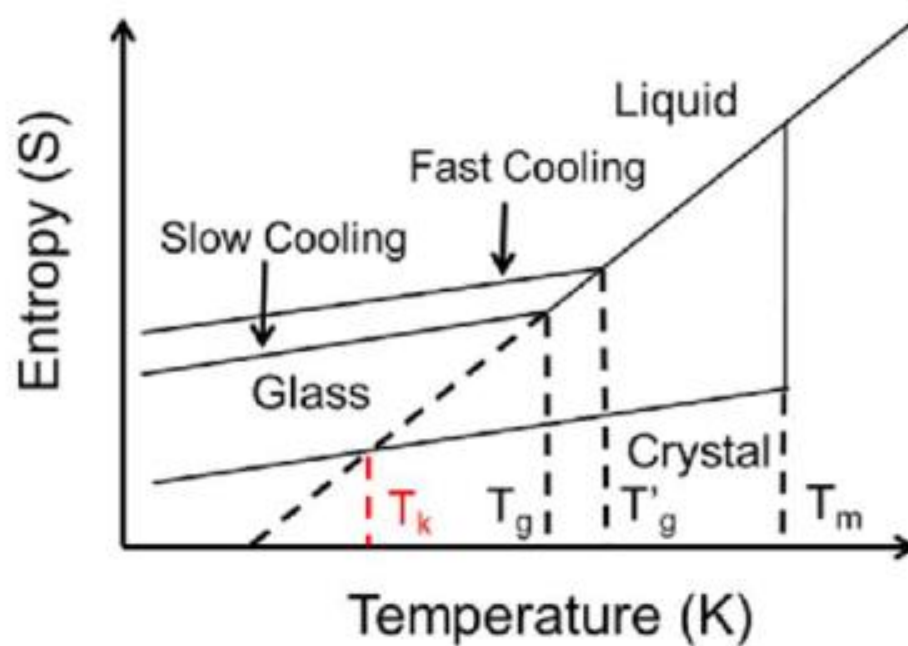
- At low temperature, it is possible that $\tau \gtrsim \tau_{\text{exp}}$
- The system is out of equilibrium and forms a **glass**
- This temperature is called **glass transition temperature** T_g
- A glass is structurally similar to a liquid, but is a different phase

Glass Transition



(Image: Jones, *Soft Condensed Matter*, Fig. 2.10)

Kauzmann Paradox

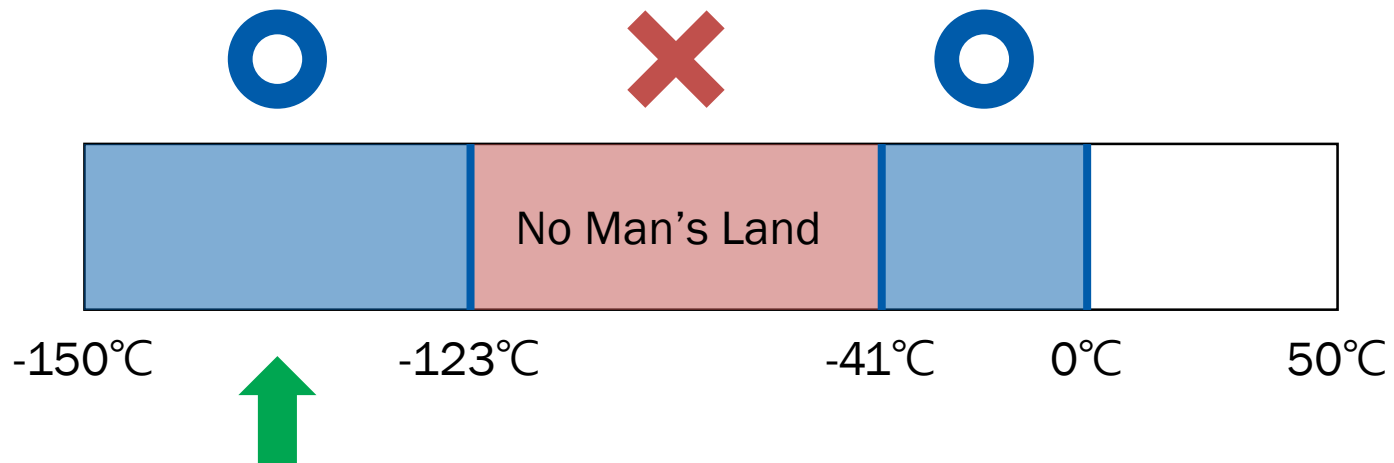


(Image: <https://doi.org/10.26434/chemrxiv-2021-66w4j-v8>)

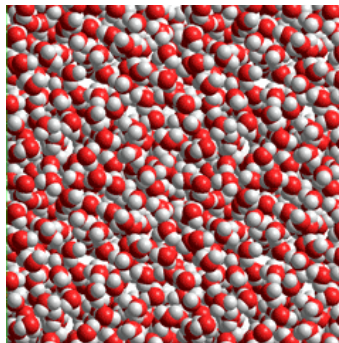
Nature of Glass Transition

- **Kinetic transition:** not a “thermodynamic” phase transition (no minimum)
- In the glassy state, the system cannot sample all the possible microstates
- Thermodynamic quantities depend on the history

Supercooling of Water



Amorphous ice



(Image: https://water.lsbu.ac.uk/water/amorphous_ice.html)

Two-Types of Amorphous Ice



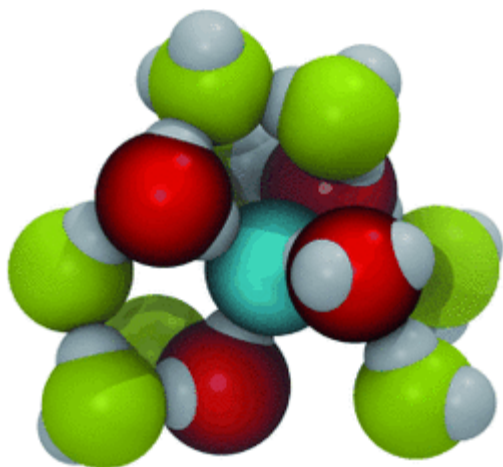
Low-density amorphous ice



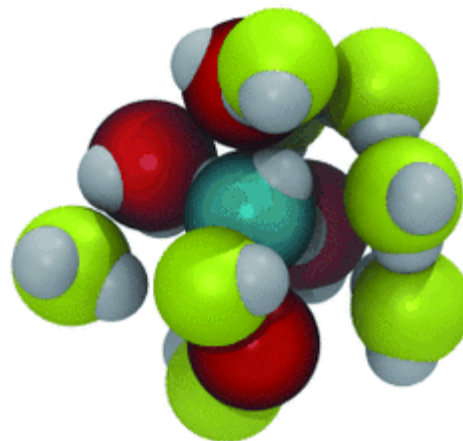
High-density amorphous ice

<https://www.youtube.com/watch?v=40e3mT9uKZw>

Structures of Amorphous Ice

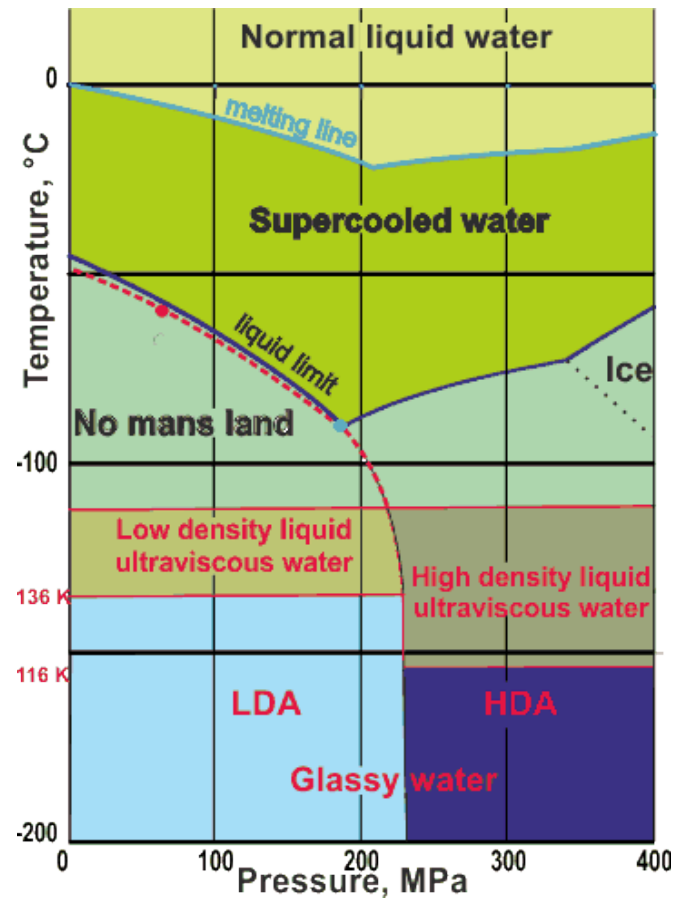


Low-density amorphous ice



High-density amorphous ice

“Metastable” Phase Diagram



(Image: https://water.lsbu.ac.uk/water/amorphous_ice.html)